



UDACITY

Data Architect Nanodegree Glossary

Course 2: Designing Data Systems

[Lesson: Foundations of Designing Systems](#)

[Lesson: Data Architecture](#)

[Lesson: Staging Data](#)

[Lesson: Operational Data Store](#)

[Lesson: Data Warehouse](#)

Lesson: Foundations of Designing Systems

Data Warehouse	A central repository of integrated data from one or more data sources that is used for reporting and data analysis.
ELT	Extract, Loading, and Transform, Technique used to move and clean the data.
ETL	Extract, Transform, and Loading. The technique used to move and clean the data
Ingest	Import data from a data source into a database
JSON	JavaScript Object Notation. Used for storing and transporting data hierarchical and semi-structured data
ODS	Operational Data Store is a transactional database to integrate and clean data from multiple data sources
Snowflake platform	Cloud-native platform for data warehousing
SQL	Structured Query Language used to manage databases in RDBMS
Workflow	A sequence of tasks, processes, or other interactions involved in processing a set of data

Lesson: Data Architecture

ACID	Database properties includes Atomicity, Consistency, Isolation, Durability designed on the foundations of relational algebra
CRUD	Create, Retrieve, Update, and Delete for transactional data management systems
Data Architecture	It is a discipline of rules, policies, and models that define what data to collect, how to use it, where to store it, and how to integrate distributed and heterogeneous systems separated in silos
Data artifacts	Data Dictionary, Data Model, Data Flows, Data Stewardship, Dimensional Model, Data Sharing Agreement
Data dictionary	A data dictionary contains information (metadata) about the elements, names, attributes, and definition related to the data in a database
Data flow diagram	Represent a flow of data through a subcomponent of, or the entirety of, a process or a system.
Data tuple	Related data elements such as a record in a database
DDL	Data Definition Language, such as creating database objects
Dimension table	table that is a descriptive key information, which contains nontransactional data used in OLAP system
Dimensional modeling	Using star schema or snowflake schemas, create a relationship among dimension and fact tables
Dimensional modeling	Using star schema or snowflake schemas, create a relationship among dimension and fact tables
Logical models	Represent business concepts within a relationship diagram
Meta database	At the highest of levels, it is a database model for metadata management.
Metadata	Data about data. Information related to the contents, format, location, size, etc
OLAP	Online analytical processing used for understanding the patterns

OLTP	On-Line Transaction Processing built on sound relational database systems
Reporting	Accessing and querying the data to find insights, trends, analysis, and other information
Row-based storage	Each row data is saved together on the disk
Silo	physical or logical separation of data without any relationship with other data sets
Single Version of Truth	Some business terms and rules have multiple meanings when there is a data dictionary. With a standard dictionary, business vocabulary can have a single version, hence data can have a single version, no matter who looks at it.
Stewardship	Accountability and responsibility for data assets and processes that ensure effective control and access.
Tribal knowledge	Undocumented information or knowledge that is known to people, who are experienced doing a particular task with any written documentation

Lesson: Staging Data

Ad Hoc loading	In emergency situations, when automation is not available, use this technique
Bulk Ingestion	Ingesting large amounts of data at intervals
Business Rules	Rules that are in place to manage access, ingestion, reporting, and updating of data
Continuous ingestion	Ingesting data in an ongoing manner, such as in real-time
Ingestion	Absorbing the incoming data at any speed at any volume
Metadata Schema	Data about data. Describes tables, columns, and business rules of all schemas
Metadatabase schema	Data about data. A schema that describes and keeps track of all the tables, columns, and business rules used in all ER models.
One Source Once Schema	In the staging area, it is easy to manage with one scheme for one source system.
Pull	In batch processing systems, data is being pulled on scheduled times
Push	In real-time streaming systems, data being pushed to downstream systems as and when it is available.
Schema	Logically group of tables, views, functions, stored procedures, and security roles
Separation of Concerns	Divide and conquer makes us think and solve data problems in a smaller context
Staging	An intermediate storage area before transforming and cleansing of data
Yottabyte	1000 Zettabytes
Zettabyte	1000 exabytes

Lesson: Operational Data Store

Anomalies of Nulls	Null values represent different meanings in data. Understanding and handling these is vital when dealing with other data sources
Anomaly	Out of character or unexpected data format, contents, timing, or size
Cleansing	Transforming and converting data from other sources into a common readable and compatible format
Column Mapping	Mapping columns from different datasets to each other
Conversions	Taking one format of data and changing it into another format. For example mm/dd/yy to dd/mm/yyyy
Decoding/encoding	Certain sensitive data such as SSN and date of birth needs to be encoded/decoded in the database
ER model	Technique to related Entities of a business domain
First Normal	Every row in the table has to be unique by creating a primary key column.
Governance	Rules, practices, and processes for management of data assets within an organization
ISO	International Standards Organization
Logical diagram	Visual diagram that represents major sources of data flow and data storage within a system
Normalization	Structuring a relational database in such a way that there is reduced data redundancy and improved data integrity. Uses first 3 forms
ODS	Operational Data Store
Reformatting	Data needs to be reformatted when putting in Word or emails

Second Normal	In addition to First normal rule, every non-prime attribute of the relationship is dependent on the whole of every candidate key
Third Normal	In addition to Second normal rule, all non-prime attributes depend only on the candidate keys and do not have a transitive dependency on another key. Why does that matter? Transitive functional dependency means that a non-key column value was determining another non-key column value.
Transformations	Changing the data from one format to another so as to provide a common format

Lesson: Data Warehouse

Accumulating Snapshot	Used to understand work-time studies of workflow steps to analyze time delays
Ad Hoc Queries	SQL database queries written dynamically as needed.
Bulk loading	Loading data in it's entirety into a table without having to go through one row at a time using ETL tools
Cloud-Native Tools	Building software taking advantage of unlimited cloud infrastructures, such as inexpensive on-demand computing and storage capacities
Columnar storage	Every column of the table runs as a separate thread and all columns run in parallel.
Conformed	A dimension, such as Date can be used for many fact tables without changing
Data Science	Extracting knowledge and patterns from various structured and unstructured data sets using scientific and statistical methods, processes, and algorithms
Data Warehouse	A central repository of integrated and cleaned data assets from one or more data sources for the purpose of reporting a single version of truth and data analysis.
Datamart	A smaller subset of the data warehouse to address a business group or a subject area
Degenerate	A primary key gives the meaning of the entire dimension without having to add any more attributes, hence the keys can be used in the fact table directly without having to create a dimension table
Denormalization	Tables that do not follow normalized rules and ER modeling relationships
Descriptive Analytics	Gives a reason for a question - What happened or what is happening
Dimension table	Tables that contain non-repetitive key values and their attributes. All dimension will have 1 to many relationships either with sub-dimension tables or Fact tables

Downstream Consumption	Interesting applications can be built when cleaned and integrated data sets are given for downstream consumption
Extraction	Pulling data from sources
Fact types	The three types are Transaction, Periodic and Accumulated
Historical Data Lineage	When every change, before and after values of every column, of a transaction is saved; this data being used for debugging the data anomalies or to find patterns.
Junk	Combination of different and unrelated attributes to avoid a large number of foreign keys in the fact table.
Operational reporting	Business operations need to know the cross-functional effectiveness of multiple integrated and cleaned transactional systems
Periodic Snapshot	A time window based snapshots can be used to collect data into data warehouse
Predictive Analytics	Takes historical data and feeds it into a machine learning model that considers key trends and patterns
Prescriptive Analytics	Business analysts and strategists come up with various courses of actions and the potential implications based on predictive analytics
RedShift	AWS cloud-native columnar storage data warehouse system built on top Postgress
Role playing	A dimension can play a different role depending upon the context.
Shrunk	Dimensions can be shrunk in order to aggregates and summaries. A quarter can be shrunk from months, months can be shrunk from dates.
Single Version of Truth	Some business terms and rules have multiple meanings when there is data dictionary. With a standard dictionary, business vocabulary can have a single version, hence data can have a single version, no matter who looks at it.
Snowflake schema	Modeling technique used for data warehousing

Snowflake schema	Elaborated technique to model complex OLAP system with many fact tables related (many to many with other fact tables and many to one with dimensional tables) much similar to ER model.
Star Schema	Fact table, which contain all details of transactions, is in the middle related (many to one) with many dimensional tables. Used to simplify OLAP complexity.
Static	Codes such as country or currency codes are static and do not change